RESOURCES AND DEVELOPMENT

Class 10 Geography - Chapter 1 | Proper AbiLearn Notes

1. Introduction to Resources

What is a Resource?

A resource is anything in our environment that can satisfy human needs, provided it is **technologically accessible**, **economically feasible**, and **culturally acceptable**. Resources are not just **natural materials**; they become resources only when **human beings use knowledge, technology, and institutions** to make them useful. For example, **river water becomes a resource** when dams, canals, pipelines, and irrigation systems help us use it for drinking, farming, and electricity.

People are the most important resource because they convert natural materials into useful assets. A forest, river, mineral deposit, or fertile land becomes valuable only when human skill, planning, and technology are used properly. Without human intervention, **nature's gifts may remain unused** and cannot fully satisfy human needs.

Resource creation depends on a relationship between **nature, technology, and institutions**. Nature provides the material, technology helps people use it, and institutions such as governments, laws, markets, and communities organise its use. This relationship supports **economic growth and social development**.

Resources are vital for human survival and development. They provide food, shelter, clothing, energy, water, land, minerals, and raw materials for industries. Without resources, human civilisation cannot exist or progress.

Historical View of Resources

In earlier times, many people believed that **resources were free gifts of nature**. Because of this belief, they used land, forests, water, minerals, and animals **indiscriminately**, without thinking about future needs or environmental limits.

This careless attitude created serious problems. People assumed nature was **inexhaustible**, but modern development proved that resources can be depleted, polluted, or degraded if used without control.

- **Depletion of resources** happened because resources were used to satisfy the **greed of a few individuals** instead of the needs of all.
- **Accumulation of resources in a few hands** divided society into **rich and poor**, increasing inequality.

- **Global ecological crises** emerged, including **global warming**, **ozone layer depletion**, **environmental pollution**, and **land degradation**.

Therefore, **sustainable development** became necessary. It means using resources carefully so that present needs are met without destroying the ability of future generations to meet their own needs.

2. Classification of Resources

Classification on the Basis of Origin

Based on origin, resources are classified into **biotic resources** and **abiotic resources**.

Biotic Resources are obtained from the **biosphere**, or the living world. They include **human beings, flora, fauna, fisheries, livestock, and forests**. These resources are renewable if they are managed carefully because living organisms can reproduce and regenerate.

Abiotic Resources are non-living resources found naturally in the environment. They include **rocks, metals, minerals, water, air, and soil**. Some abiotic resources such as water and air may be renewable, while others such as minerals and fossil fuels are non-renewable.

Classification on the Basis of Exhaustibility

Based on exhaustibility, resources are classified into **renewable resources** and **non-renewable resources**.

Renewable Resources can be replenished naturally within a relatively short period. Examples include **water, forests, wind, solar energy, and biomass**. However, even renewable resources can become scarce if they are overused or polluted.

Non-Renewable Resources take millions of years to form and cannot be replaced quickly once exhausted. Examples include **coal, petroleum, natural gas, iron ore, copper, and bauxite**. These resources must be used very carefully because they are limited.

Recyclable resources, such as metals, can be reused after recycling. Recycling scrap iron, aluminium, copper, and other metals reduces mining pressure and extends resource availability.

Classification on the Basis of Ownership

Based on ownership, resources are divided into **individual, community, national, and international resources**.

Individual Resources are privately owned by individuals. Examples include **farmland, houses, cars, wells, and books**. Their use is protected by law but is still subject to public regulations.

Community Resources are owned and used collectively by a community. Examples include **public parks, playgrounds, grazing grounds, burial grounds, village ponds, and common forests**. These resources serve the welfare of the whole community.

National Resources belong to the nation and are managed by the government. Examples include **roads, canals, railways,**

minerals, forests, wildlife, and water resources. The state can acquire private land for public purposes such as roads, schools, dams, or hospitals.

International Resources are regulated by international institutions. Oceanic resources beyond **200 nautical miles** of the Exclusive Economic Zone, Antarctica, and outer space are examples. No single country can claim exclusive ownership over them.

Classification on the Basis of Status of Development

Based on development status, resources are classified as **potential resources, developed resources, stock, and reserves**.

Potential Resources exist in a region but have not yet been fully used. For example, **solar and wind energy potential** in many areas remains underutilised.

Developed Resources have been surveyed, measured, and prepared for use. Their **quality and quantity** are known, and they can be used with available technology.

Stock refers to materials that exist but cannot currently be used because we do not have suitable technology. For example, hydrogen and oxygen in water can produce energy, but extracting them efficiently at large scale is still difficult.

Reserves are a part of stock that can be used with existing technology but are saved for the future. For example, water stored in dams is a reserve for irrigation and hydroelectric power.

3. Resource Development

What is Resource Development?

Resource development is the process of transforming resources into useful forms through human activities. It includes **exploration, extraction, processing, conservation, and proper utilisation** of resources.

A resource is developed only when it can be used effectively. For example, minerals must be located, mined, transported, processed, and converted into useful goods. Similarly, land becomes productive when it is irrigated, fertilised, protected from erosion, and cultivated properly.

- **Technology availability** decides whether resources can be extracted and processed.
- **Economic feasibility** decides whether development is cost-effective.
- **Infrastructure** such as roads, electricity, ports, and storage supports development.
- **Skilled labour** and education improve resource use.
- **Capital investment** is needed for large development projects.

Human activities such as **agriculture, industry, transport, trade, mining, and services** transform resources into useful goods and services.

Factors Influencing Resource Development

Nature provides raw materials, but **technology makes them usable**. Crude oil, uranium, iron ore, and even river water become valuable only when technology allows us to use them properly.

Institutions also control resource development. Government policies, laws, property rights, environmental rules, and local communities determine how resources are used and protected.

People are the key agents of resource development. Their education, skills, innovation, discipline, and planning decide whether resources are wasted or converted into long-term assets.

4. Resource Planning in India

Need for Resource Planning

Resource planning means identifying, evaluating, and using resources in a systematic and sustainable way. It is necessary because resources are limited, unevenly distributed, and under pressure from population growth and development.

India has **great regional diversity** in resource availability. Some regions are rich in minerals but lack infrastructure, while others have good energy potential but face water scarcity. Therefore, planning is needed at the **national, state, regional, and local levels**.

- **Jharkhand, Chhattisgarh, and Madhya Pradesh** are rich in minerals and coal but often lack sufficient infrastructure.
- **Rajasthan** has high solar and wind energy potential but suffers from water scarcity.
- **Himalayan regions** have water resources but difficult terrain limits development.
- **Coastal regions** have marine resources but face pollution and overfishing.

Resource planning helps ensure **balanced regional development**, reduces wastage, avoids conflicts, and integrates economic growth with environmental protection.

Three-Stage Process of Resource Planning

Stage 1: Identification and Listing of Resources involves surveying, mapping, and estimating the quantity and quality of resources. Modern tools such as **remote sensing, GPS, and GIS** help create accurate resource databases.

Stage 2: Establishing a Planning System means developing suitable technology, building skills, creating institutions, arranging capital, and preparing mechanisms to use resources effectively.

Stage 3: Aligning with National Development Goals ensures that resource development supports poverty reduction, regional balance, environmental protection, and overall national progress.

5. Land Resources

Land as a Resource

Land is one of the most important natural resources because all human activities take place on land or depend on land. It supports settlement, agriculture, forests, industries, roads, railways, grazing, mining, and ecological systems.

- **Forests** provide timber, fuel, biodiversity, rainfall support, and ecological balance.
- **Agriculture** uses land to grow crops and raise livestock.
- **Grazing grounds** support cattle and other animals.
- **Built-up areas** are used for houses, industries, schools, roads, and infrastructure.
- **Barren and waste land** is currently less productive but may be improved with effort.
- **Non-agricultural land uses** include roads, railways, canals, airports, and factories.

Land is a **finite resource**. Its area cannot be increased, but its productivity can be improved through proper management, irrigation, soil conservation, and sustainable planning.

Land Utilisation in India

India's land use pattern has changed due to population growth, agriculture, urbanisation, industrialisation, and infrastructure development.

- **Net Sown Area** is about 43% of total land area and is used for agriculture.
- **Forest Area** is about 23%, while the National Forest Policy recommends 33%.
- **Grazing Grounds** cover about 4%.
- **Barren and Waste Land** is about 6%.
- **Land Put to Non-Agricultural Uses** is about 8%.
- **Culturable Waste Land** is about 5%.
- **Fallow Land** is about 7%, left uncultivated temporarily to regain fertility.

Net sown area varies widely. **Punjab and Haryana** have very high net sown area, while **Arunachal Pradesh, Mizoram, and Andaman-Nicobar Islands** have very low net sown area due to forests, hills, and terrain.

Forest area is below the desired target in many states because of agriculture, mining, urban expansion, and infrastructure growth.

Factors Affecting Land Use

Land use is controlled by both physical and human factors.

- **Physical factors** include topography, climate, soil type, water availability, slope, and natural vegetation.
- **Human factors** include population density, technology, economic development, culture, traditions, and government policy.

Plains are generally suitable for agriculture and settlements, while mountains are more suitable for forestry, tourism, and hydropower. Industrialisation and urbanisation often convert agricultural land into non-agricultural land.

6. Soil as a Resource

Soil - The Most Important Renewable Natural Resource

Soil is the most important renewable natural resource because it is the medium for plant growth. It supports agriculture, forests, grasslands, microorganisms, animals, and human life.

Soil is formed by the breakdown of rocks over a very long time through **weathering**. Soil formation depends on parent rock, climate, relief, vegetation, organisms, and time.

- **Parent rock** affects mineral content.
- **Climate** controls weathering and leaching.
- **Relief** affects erosion and deposition.
- **Vegetation** adds organic matter.
- **Time** is necessary because soil formation is slow.

Soil contains **mineral matter, organic matter, water, and air**. **Humus**, the decomposed organic matter, improves fertility and water-holding capacity.

Classification of Soils in India

Indian soils are classified based on formation, colour, thickness, texture, age, chemical properties, and physical properties. The major types are **alluvial, black, red and yellow, laterite, arid, and forest soils**.

1. Alluvial Soil

Alluvial soil is the most widespread and important soil in India. It covers large parts of the northern plains, river valleys, and coastal deltas. It is deposited by rivers such as the Indus, Ganga, and Brahmaputra.

- It is **highly fertile** and suitable for wheat, rice, sugarcane, cotton, pulses, and oilseeds.
- It is generally rich in **potash** but may be poor in phosphorus.
- **Khadar** is newer alluvium, deposited by floods and more fertile.
- **Bhangar** is older alluvium, found on higher terraces and may contain kankar nodules.

2. Black Soil (Regur Soil)

Black soil, also called **Regur soil**, is ideal for cotton cultivation and is therefore known as **cotton soil**. It is formed from volcanic rocks and is common in the Deccan Plateau.

- It has excellent **moisture-retaining capacity**.
- It is rich in calcium carbonate, magnesium, potash, and lime.
- It is poor in phosphorus and nitrogen.
- It develops deep cracks in hot weather, helping aeration.

- It is sticky when wet and hard when dry.

It is found in Maharashtra, Gujarat, Madhya Pradesh, parts of Chhattisgarh, Andhra Pradesh, Karnataka, and Tamil Nadu. It supports cotton, sugarcane, wheat, jowar, and millet.

3. Red and Yellow Soil

Red and yellow soil forms in areas of crystalline and metamorphic rocks. The red colour is due to iron diffusion, while yellow colour appears when iron is hydrated.

- It is generally poor in nitrogen, phosphorus, and lime.
- It is found in the eastern and southern Deccan Plateau, Odisha, Chhattisgarh, southern Maharashtra, Tamil Nadu, Karnataka, and some Himalayan foothills.
- It can support millet, pulses, oilseeds, and tobacco with irrigation and fertilisers.

4. Laterite Soil

Laterite soil forms in areas of high temperature and heavy rainfall. Heavy rainfall causes **leaching**, removing lime and silica and leaving iron and aluminium compounds behind.

- It is poor in nitrogen, phosphorus, potassium, and organic matter.
- It becomes hard like brick when dry.
- It is found in Karnataka, Kerala, Tamil Nadu, Madhya Pradesh, Odisha, Assam, and northeastern states.
- With manure and fertilisers, it can support tea, coffee, rubber, and cashew.

5. Arid and Desert Soil

Arid soil is found in dry regions with low rainfall and high evaporation. It is common in Rajasthan and parts of Gujarat, Haryana, and Punjab.

- It is sandy, saline, and low in humus and moisture.
- A kankar layer may restrict water infiltration.
- With irrigation, it can support drought-resistant crops such as bajra, jowar, and millet, and also wheat and cotton in some areas.

6. Forest Soil

Forest soil is found in hilly and mountainous regions under forest cover. Its character changes according to relief, climate, and vegetation.

- It is usually rich in organic matter but may be acidic and nutrient-poor because of leaching.
- It is loamy and silty in valleys but coarse on upper slopes.
- It is found in the Himalayas, northeastern states, Western Ghats, and Eastern Ghats.
- It supports tea, coffee, spices, and tropical fruits in suitable regions.

7. Soil Erosion and Soil Conservation

What is Soil Erosion?

Soil erosion is the removal of the top fertile layer of soil by water, wind, gravity, glaciers, or human activities. Since topsoil contains humus and nutrients, its loss reduces soil fertility and agricultural productivity.

- It causes loss of fertility and crop productivity.
- It leads to siltation of rivers, reservoirs, and canals.
- It increases flooding and reduces water absorption.
- It causes land degradation and desertification.
- It pollutes water bodies with sediment.

Causes of Soil Erosion

Natural causes include **water, wind, gravity, and glaciers**. Human causes are often more serious because they remove vegetation and expose soil.

- **Deforestation** exposes soil to rain and wind.
- **Overgrazing** removes grass cover.
- **Improper farming** such as ploughing along slopes increases runoff.
- **Mining and quarrying** disturb land and vegetation.
- **Construction** removes topsoil and changes drainage.
- **Jhum cultivation** clears forest patches and may increase erosion.

Types of Soil Erosion

Sheet erosion happens when water flows as a thin sheet and removes topsoil from a wide area. It is difficult to notice but damaging.

Rill erosion forms small channels or grooves due to flowing water. It is an early stage of more serious erosion.

Gully erosion creates deep channels that destroy farmland and produce badland topography. It is severe and difficult to reverse.

Wind erosion occurs in dry regions where loose soil is blown away by strong winds. It is common in arid and semi-arid areas.

Methods of Soil Conservation

- **Contour Ploughing** means ploughing along contour lines to slow water flow.
- **Terrace Farming** creates steps on slopes to reduce erosion in hilly areas.
- **Strip Cropping** grows crops in alternate strips with grass to reduce wind force.
- **Shelter Belts** are rows of trees planted to reduce wind speed.
- **Afforestation and Reforestation** increase tree cover and hold soil.
- **Controlled Grazing** prevents overgrazing and allows vegetation to recover.
- **Check Dams** slow water flow and trap sediment in gullies.

- **Crop Rotation** maintains soil fertility and improves soil structure.
- **Mulching** protects soil from erosion and adds organic matter.

8. Conservation of Resources

Why Conservation is Necessary

Resource conservation is necessary because resources are limited, population is increasing, and development is raising demand. If resources are wasted today, future generations will face scarcity and environmental damage.

Conservation supports **sustainable development**. It protects ecological balance, reduces pollution, saves money, and ensures that resources remain available for future needs.

Principles of Resource Conservation

- **Reduce** resource use by avoiding wastage of water, electricity, fuel, paper, and raw materials.
- **Reuse** items such as bags, bottles, containers, and old materials instead of throwing them away.
- **Recycle** paper, glass, plastic, and metals to reduce pressure on new resources.
- **Replace** non-renewable resources with renewable alternatives such as solar and wind energy.

Mahatma Gandhi's View on Conservation

Mahatma Gandhi said: "There is enough for everybody's need, but not for anybody's greed." This means resources are sufficient if used responsibly, but greed leads to depletion, inequality, and environmental crisis.

Gandhi supported **simple living, responsible consumption, and wise use of resources**. His idea is directly connected to modern sustainable development.